Supplementary Information

A Coupled Chemo-Mechanical and Life-Cycle Framework for Oyster-Shell-Derived Cementitious Materials in Aggressive Infrastructure Environments

Tables S1–S8 give the calibration, validation and re-evaluation details summarised in Sections 4.4, 6.4, 7.4 and 7.5 of the manuscript. All numbers are generated by the scripts in the repository https://github.com/sandlerleon/oyster-shell-concrete-mc.

Table S1. Calibration set-up (scallop-shell data)

Item	Value
Dataset	Abbas and Kudukkan (2025), Constr. Mater. 5(4):82, binder 350 kg/m3, water 160 kg/m3, 28-day strength
Mixes (shell-derived powder)	S0F15 50 kg/m3 (r=0.143, d=90 um); S15F15 95 (r=0.271, d=64); S30F15 140 (r=0.400, d=55)
Observed relative strength	S0F15 1.00 (text); S15F15 0.865 (derived from moduli, E ~ sqrt(fc)); S30F15 0.75 (text)
Extension	effective strength-efficiency coefficient k_f; cement volume in the gel/space ratio multiplied by max(1 + k_f r/(1 - r), 0)
Algorithm	rejection ABC; 300,000 draws per case; tolerance 0.07 on S0F15 and S30F15; S15F15 held out; accepted 10999 (A), 10975 (B), 10999 (C)
Original model (k_f = 0)	predicted relative strength: 0.79, 0.61, 0.43; none of 100,000 draws within tolerance

Table S2. Prior and posterior parameter summaries, Abbas-only calibration (case A draws)

Parameter	Prior median	Posterior median	Posterior P10	Posterior P90
kf	0.599	0.648	0.537	0.839
nu	0.75	0.71	0.13	1.34
bpack	0.15	0.169	0.0378	0.276
o_org	0.0213	0.0218	0.0128	0.0322
eta_w	0.6	0.579	0.354	0.827
kap_o	31.6	41.3	7.92	153
kap_s	2.4	2.31	1.89	2.84
kimp	0.025	0.0248	0.00478	0.045
k0	0.26	0.259	0.2	0.334
acap	0.935	0.936	0.907	0.963
S0	78	78.1	68.7	88.9

Table S3. Independent oyster-shell data: observed and predicted relative strength (frozen Abbas posterior median with 5-95% interval; original model; k_f = 0.25; Bolomey dilution)

Study	Mix	r	Observed	Calibrated frozen	k_f = 0	k_f = 0.25	Bolomey
KWON	OSP15	0.150	0.652	0.94 (0.93-0.96)	0.79	0.85	0.80
KWON	OSP30	0.300	0.429	0.87 (0.86-0.89)	0.57	0.68	0.60
ABIN	r2.5	0.025	1.053	0.99 (0.99-1.02)	0.97	0.98	0.97
ABIN	r5	0.050	1.105	0.99 (0.98-1.01)	0.94	0.96	0.94
ABIN	r7.5	0.075	1.053	0.98 (0.97-1.00)	0.91	0.94	0.91
ABIN	r10	0.100	1.039	0.97 (0.96-0.99)	0.87	0.91	0.88
MEX	0.14 mm	0.100	1.037	0.95 (0.94-0.95)	0.83	0.88	0.85
MEX	0.17 mm	0.100	0.978	0.94 (0.94-0.95)	0.83	0.87	0.85
MEX	0.25 mm	0.100	1.000	0.94 (0.94-0.95)	0.83	0.87	0.85
MEX	0.42 mm	0.100	1.015	0.94 (0.93-0.95)	0.83	0.87	0.85
MEX	0.84 mm	0.100	0.882	0.94 (0.93-0.95)	0.83	0.87	0.85

Data sources and assumptions: KWON = Kwon and Wang 2024 (Buildings 14:1578, Tables 1 and 3; binder 568.16 kg/m3, w/b 0.50, size assumed 30 um). ABIN = Abinaya et al. as transcribed in Kim and Lee 2023 (Buildings 13:3078, Table A1; secondary; binder 400 kg/m3 and size 30 um assumed). MEX = De Dios-Suarez et al. 2026 (Materials 19:1656, Table 2; mortar 1:4, w/c 0.68; 10 wt% of cement interpreted as replacement; sieve aperture as particle size).

Table S4. Study-level k_f (rejection ABC, prior U(-1.5, 1.5), tolerance 0.07), random-effects predictive distribution and leave-one-study-out results

Study	k_f median (P10 to P90)	Accepted draws	Leave-one-out MAE / bias
ABBAS	0.65 (0.54 to 0.85)	4414	0.038 / -0.033
KWON	-0.40 (-0.65 to -0.17)	23638	0.460 / +0.460
ABIN	1.36 (1.15 to 1.47)	17990	0.108 / -0.108
MEX	0.76 (0.69 to 1.04)	31	0.077 / -0.064

Random-effects predictive: Student-t with location 0.594, scale 0.817, 3 degrees of freedom, truncated to [-1, 1.5]. Size-dependent formulation: no draw reproduces the Abbas mixes (best-fit k_f 0.70) or the 2026 mortar data (best-fit 1.50).

Table S5. Primary (pooled) results for P1, 30 um shell: carbon change, life difference and joint-advantage probability

Case	Design	dE % P50 (P10 to P90)	fc28 vs OPC	dL yr P50 (P10 to P90)	P(non-inferior)	P(adv, mat)
A	10% same w/b	-8.5 (-8.8 to -8.1)	0.95	-2.7 (-11.3 to +1.2)	0.18	0.18
A	10% matched	-6.2 (-8.7 to +5.1)	1.02	-0.1 (-6.7 to +9.0)	0.48	0.15
A	25% same w/b	-21.2 (-22.0 to -20.4)	0.87	-8.4 (-33.5 to +2.6)	0.14	0.14
A	25% matched	-15.0 (-21.6 to	1.02	-1.8 (-20.6 to	0.41	0.21

		+9.7)		+15.8)		
A	50% same w/b	-42.5 (-44.0 to -40.7)	0.73	-16.6 (-53.5 to +1.1)	0.11	0.11
A	50% matched	-37.0 (-43.3 to +13.0)	1.02	-8.2 (-43.0 to +16.8)	0.30	0.19
B	10% same w/b	-8.1 (-8.4 to -7.7)	0.94	+0.0 (+0.0 to +0.0)	0.55	0.55
B	10% matched	-5.7 (-8.3 to +7.6)	1.03	+0.0 (+0.0 to +0.0)	0.77	0.33
B	25% same w/b	-20.2 (-21.1 to -19.2)	0.85	+0.0 (-5.7 to +0.0)	0.05	0.05
B	25% matched	-14.2 (-20.7 to +22.8)	1.04	+0.0 (+0.0 to +0.0)	0.46	0.14
B	50% same w/b	-40.5 (-42.2 to -38.5)	0.68	-18.9 (-81.5 to +0.0)	0.00	0.00
B	50% matched	-34.6 (-41.5 to +26.7)	1.03	+0.0 (-68.2 to +0.0)	0.29	0.06
C	10% same w/b	-8.5 (-8.8 to -8.2)	0.95	-2.5 (-7.7 to -0.8)	0.00	0.00
C	10% matched	-6.2 (-8.7 to +5.1)	1.02	-1.0 (-5.1 to +4.2)	0.33	0.00
C	25% same w/b	-21.3 (-22.1 to -20.5)	0.87	-5.5 (-16.9 to -1.9)	0.00	0.00
C	25% matched	-15.1 (-21.7 to +9.8)	1.02	-2.3 (-11.0 to +10.8)	0.33	0.03
C	50% same w/b	-42.7 (-44.2 to -41.0)	0.73	-10.8 (-28.7 to -3.5)	0.00	0.00
C	50% matched	-37.2 (-43.5 to +13.1)	1.02	-6.2 (-23.2 to +11.2)	0.21	0.04

Table S6. Blended-cement benchmark (50% slag, Life-365 treatment, screening)

Case	Configuration	Carbon change vs reference (%, median)	Life (yr) / life difference	OPC life / P(adv)
pavement	blend50	-42.2	96.2	24.4
pavement	blend30	-25.3	50.3	24.4
pavement	blend50_slag12	-39.5	96.2	24.4
pavement	shell P1_r25u_vs_blend50	+36.3	dL -54.8	P(adv) 0.000
pavement	shell P1_r25m_vs_blend50	+47.4	dL -40.2	P(adv) 0.000
pavement	shell P1_r50m_vs_blend50	+10.3	dL -49.2	P(adv) 0.000
pavement	shell P1_r25m_vs_blend30	+14.0	dL -20.6	P(adv) 0.000
dam	blend50	-40.3	100.0	100.0
dam	blend30	-24.2	100.0	100.0
dam	blend50_slag12	-37.7	100.0	100.0
dam	shell P1_r25u_vs_blend50	+33.5	dL +0.0	P(adv) 0.000
dam	shell P1_r25m_vs_blend50	+43.6	dL +0.0	P(adv) 0.000
dam	shell P1_r50m_vs_blend50	+9.4	dL +0.0	P(adv) 0.000
dam	shell P1_r25m_vs_blend30	+13.1	dL +0.0	P(adv) 0.000
culvert	blend50	-42.4	68.2	18.2

culvert	blend30	-25.5	34.5	18.2
culvert	blend50_slag12	-39.7	68.2	18.2
culvert	shell P1_r25u_vs_blend50	+36.6	dL -53.6	P(adv) 0.000
culvert	shell P1_r25m_vs_blend50	+47.8	dL -42.3	P(adv) 0.000
culvert	shell P1_r50m_vs_blend50	+10.3	dL -48.2	P(adv) 0.000
culvert	shell P1_r25m_vs_blend30	+14.1	dL -16.0	P(adv) 0.000

Table S7. Alternative k_f formulations: P(adv, mat) and median life difference

Case	Design	P(adv, mat) strength-only / coupled	dL P50 strength-only / coupled
A	r25u	0.14 / 0.33	-8.4 / -3.7
A	r25m	0.21 / 0.45	-1.8 / +1.8
A	r50m	0.19 / 0.40	-8.2 / +0.0
B	r25u	0.05 / 0.20	+0.0 / +0.0
B	r25m	0.14 / 0.32	+0.0 / +0.0
B	r50m	0.06 / 0.17	+0.0 / +0.0
C	r25u	0.00 / 0.18	-5.5 / -3.6
C	r25m	0.03 / 0.29	-2.3 / +0.2
C	r50m	0.04 / 0.26	-6.2 / -1.8

Table S8. Carbon check against the calibration paper

Item	Value
Control embodied carbon (paper's factors)	331.0 kg CO2e/m3
S15F15 / S30F15 change vs control	-24.4% / -36.0%
Model shell emission factor at 55 um (P10 / P50 / P90)	0.036 / 0.058 / 0.084 kg CO2e/kg
Laboratory value in the paper	0.059 kg CO2e/kg (system boundary not stated)

Table S9. Leave-one-study-out analysis of the P1 results at r = 25% with strength matching (pooled k_f distribution of the remaining studies)

Left out	Case	k_f location / scale	dE % (P10 to P90)	dL yr (P10 to P90)	P(adv, mat)
ABBAS	A	0.57 / 1.03 (df 2)	-14.5 (-21.6 to +11.0)	-2.5 (-22.5 to +15.3)	0.19
ABBAS	B	0.57 / 1.03 (df 2)	-13.8 (-20.7 to +28.5)	+0.0 (+0.0 to +0.0)	0.11
ABBAS	C	0.57 / 1.03 (df 2)	-14.6 (-21.7 to +11.0)	-2.0 (-10.8 to +14.2)	0.02
KWON	A	0.92 / 0.44 (df 2)	-20.4 (-21.8 to -2.4)	-3.5 (-20.8 to +11.5)	0.24

KWON	B	0.92 / 0.44 (df 2)	-19.2 (-20.8 to -1.7)	+0.0 (+0.0 to +0.0)	0.13
KWON	C	0.92 / 0.44 (df 2)	-20.5 (-21.9 to -2.4)	-3.9 (-13.8 to +1.2)	0.02
ABIN	A	0.34 / 0.74 (df 2)	-9.2 (-21.5 to +10.9)	-1.0 (-19.2 to +16.3)	0.19
ABIN	B	0.34 / 0.74 (df 2)	-8.6 (-20.4 to +27.1)	+0.0 (+0.0 to +0.0)	0.15
ABIN	C	0.34 / 0.74 (df 2)	-9.2 (-21.6 to +11.0)	-1.1 (-9.5 to +14.8)	0.03
MEX	A	0.54 / 1.02 (df 2)	-13.7 (-21.6 to +11.1)	-2.4 (-22.5 to +15.3)	0.18
MEX	B	0.54 / 1.02 (df 2)	-13.1 (-20.7 to +28.7)	+0.0 (+0.0 to +0.0)	0.11
MEX	C	0.54 / 1.02 (df 2)	-13.8 (-21.7 to +11.2)	-1.9 (-10.8 to +14.4)	0.02
none (primary)	A	0.59 / 0.82 (df 3)	-15.0 (-21.6 to +9.7)	-1.8 (-20.6 to +15.8)	0.21
none (primary)	B	0.59 / 0.82 (df 3)	-14.2 (-20.7 to +22.8)	+0.0 (+0.0 to +0.0)	0.14
none (primary)	C	0.59 / 0.82 (df 3)	-15.1 (-21.7 to +9.8)	-2.3 (-11.0 to +10.8)	0.03

Table S10. Joint-advantage probability under fixed transport models (same w/b / strength-matched, r = 25%)

Transport model	Pavement	Dam	Culvert
chi0.0|strength_only	0.17 / 0.26	0.17 / 0.25	0.00 / 0.05
chi0.0|coupled	0.54 / 0.68	0.54 / 0.63	0.43 / 0.66
chi0.5|strength_only	0.14 / 0.21	0.02 / 0.12	0.00 / 0.02
chi0.5|coupled	0.33 / 0.48	0.13 / 0.28	0.14 / 0.23
chi1.0|strength_only	0.14 / 0.18	0.00 / 0.06	0.00 / 0.01
chi1.0|coupled	0.14 / 0.18	0.01 / 0.07	0.01 / 0.02

Table S11. Study-level prediction errors (11 independent points)

Model	n	MAE	RMSE	Bias	MAE Kwon	MAE Abinaya	MAE mortar 2026
Abbas-calibrated posterior (frozen)	11	0.124	0.174	+0.019	0.368	0.080	0.063
original inert-filler model (k_f = 0)	11	0.145	0.151	-0.094	0.140	0.139	0.152
k-value rule in the same model (k_f = 0.25)	11	0.133	0.146	-0.051	0.225	0.116	0.110
Bolomey dilution only	11	0.141	0.147	-0.083	0.159	0.141	0.134
Bolomey k-value 0.25	11	0.131	0.146	-0.045	0.234	0.121	0.098